

Toward a Unified Database of Hybrid Broadband Ground-Motion Simulations for Historical Mw 3.5-7.8 Crustal and Subduction Earthquakes in New Zealand

Felipe Kuncar, Brendon Bradley, Robin Lee, Cesar Pajaro, Michael Dupuis, Jake Faulker, Sung Bae
University of Canterbury, Department of Civil and Environmental Engineering | felipe.kuncar@canterbury.ac.nz

1. Introduction

Hybrid broadband ground-motion simulation can be used in seismic hazard and response-history analyses, offering significant potential advantages over empirical models currently used in practice to predict earthquake-induced ground motion. However, validation against observations from historical events is necessary to build confidence in the predictive capabilities of simulations and identify areas for modelling improvements.

2. Recent Validation Studies in New Zealand

Several validation studies have been conducted in New Zealand (NZ), using the Graves and Pitarka (2010, 2015, 2016) hybrid simulation method (GP method) and large datasets of recordings (Figure 1).

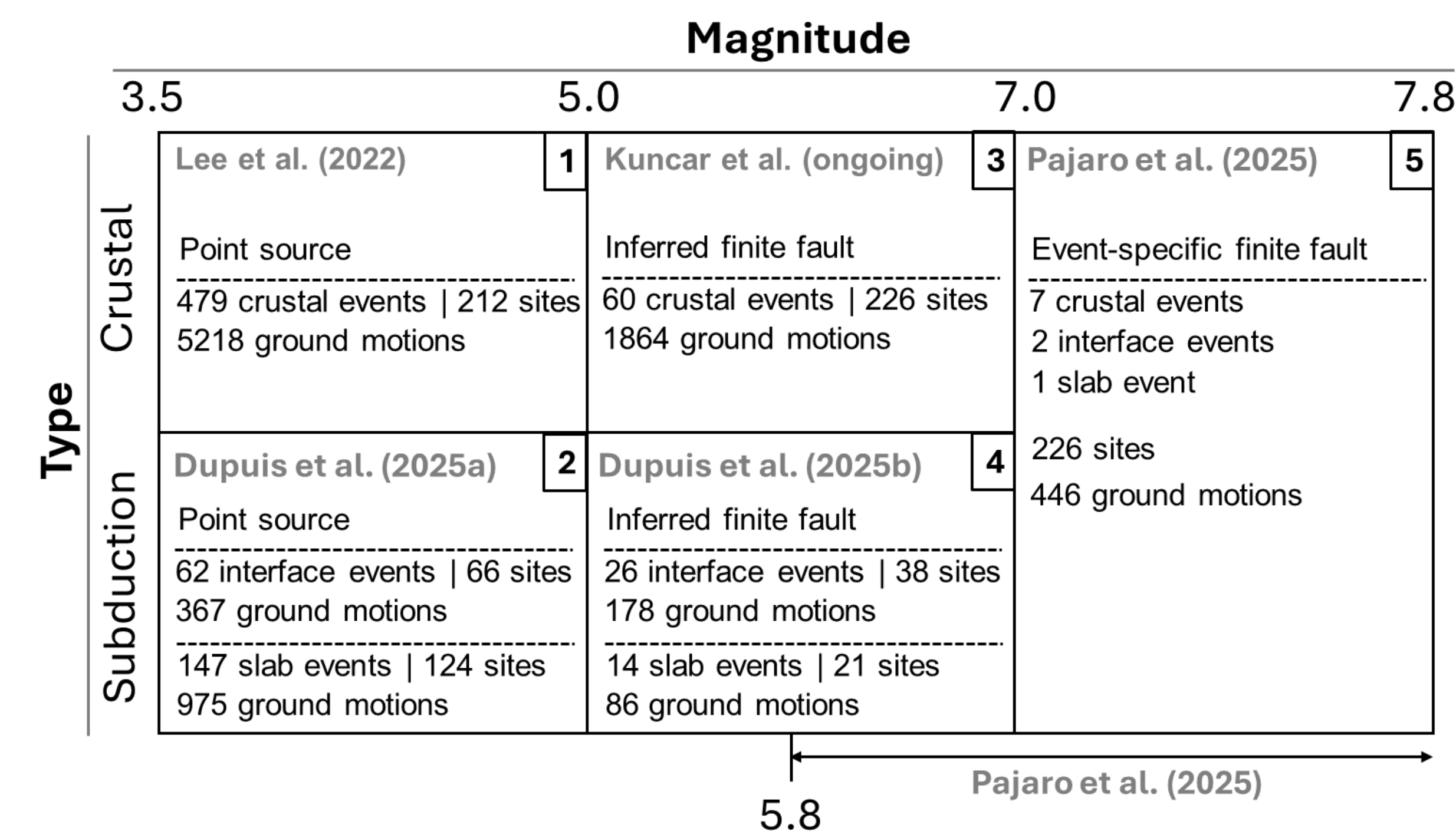


Figure 1: Main validation studies in NZ using large recording datasets, spanning small to large magnitudes and both crustal and subduction events.

While these individual studies have provided valuable insights and improvements in simulations based on subsets of the available observational data, a large-scale integrated dataset is needed to fully assess and enhance simulation performance across all earthquake types and site conditions. For example, Table 1 illustrates that different criteria and sources of information were considered in each validation study. An integrated database requires unified considerations.

Table 1: Characteristics of main validation studies in New Zealand.

Study in Fig. 1	Computational features	Site amp. model	NZ velocity model version	Observational database
1	$V_{s,min} = 500 \text{ m/s}$ $h_{grid} = 100 \text{ m}$ $f_t = 1 \text{ Hz}$	CB14	2.02	Study-specific
2		CB14	2.07	NZ GMDB v3.4
3		BA18	2.09	NZ GMDB v4.3
4		CB14	2.07	NZ GMDB v3.4
5		CB14	2.07	NZ GMDB v4.3

3. Prior validation studies in other regions

Globally, similar efforts have been undertaken, primarily focused on small-to-moderate crustal earthquakes (Table 2). These studies have been conducted considering different simulation methods and frequency ranges.

Table 2: Global validation studies.

Study	Region	Magnitude	No. of events	No. of recordings	f_{max} (Hz)
Goulet et al. (2015); Dreger et al. (2015)	USA (WUS, CENA), JPN	5.4 – 7.2	12	394	Hybrid $f_t=1 \text{ Hz}$
Taborda et al. (2016)	Southern California	3.6 – 5.4	30	3071	1
Paolucci et al. (2021); BB-SPEEDset (2025)	ITA, GRC, JPN, TUR, FRA, CHN	4.9 – 7.0	11	~127	Hybrid $f_t=0.5-1.3 \text{ Hz}$
Graves (2022)	California	5.9 – 7.3	12	504	Hybrid $f_t=1 \text{ Hz}$
Sajan et al. (2024, ongoing)	Southern California	4.0 – 5.5	27	2917	1
Pinilla-Ramos et al. (2025)	San Francisco	3.8 – 4.4	7	943	5
Sajan et al. (2025, ongoing)	Northern California	4.0 – 5.5	59	7416	1
This study (ongoing)	New Zealand	3.5 – 7.8	~849	~12,751	Hybrid $f_t=1 \text{ Hz}$

4. Selected ground motions for a unified database

For the proposed database, the following screening criteria are considered (using the latest version of the NZ ground-motion database, NZGMDB, v4.3):

- Only Crustal, Slab, and Interface earthquakes
- Earthquakes with a CMT solution and $M_W \geq 3.5$
- $R_{rup} \leq 300 \text{ km}$ and usable period of at least 2 s
- At least 3 recordings per earthquake and station

Based on these criteria, a total of 44,379 ground motions from 1,503 earthquakes, recorded at 369 strong-motion stations, are available. Figure 2 shows the spatial distribution of these events and stations.

Based on the available waveform properties and the planned geographic extent of the simulation domains, we are provisionally considering 12,751 ground motions from 849 events, recorded at 348 stations. This represents a substantially larger validation dataset than those used in previous studies (e.g., Figure 1; Table 2).

5. Simulations and Expected Outputs

Table 3 summarizes the main features of the simulations that will be performed for the integrated database, using the GP method. The observational database to be used will be the NZGMDB v4.4, currently under development, which will potentially increase the amount of available validation data.

In addition to paired observed and simulated waveforms, this open-source database will include several intensity measures of engineering interest, computed for the simulations, the corresponding observations, and empirical models. For response spectra, the empirical models used in the 2022 NZ National Seismic Hazard model (Bradley et al., 2024) will be considered. All relevant simulation and empirical input parameters will be provided.

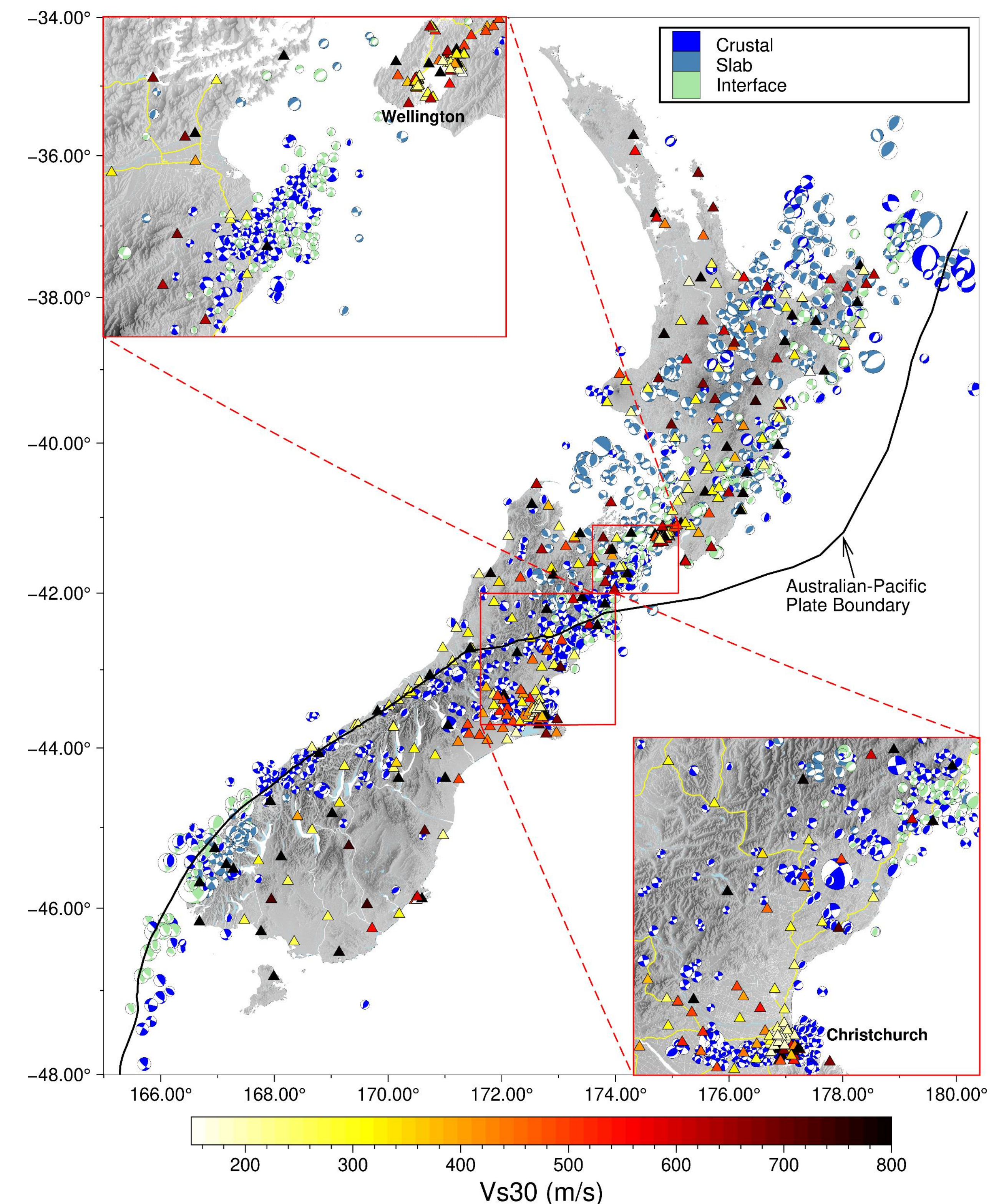
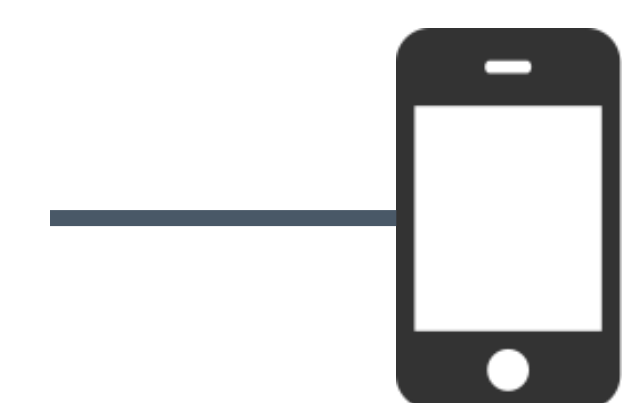


Figure 2: Location of potential stations and earthquake sources to be considered.

Table 3: Features of simulations to be performed.

Minimum shear-wave velocity ($V_{s,min}$)	500 m/s
Grid spacing (h_{grid})	100 m
Transition frequency (f_t)	1 Hz
Site amplification model	BA18
Velocity model	NZVM v2.09
Input parameters (e.g., κ_0 , Brune stress parameter, rupture velocity)	Common parametrization for each tectonic class

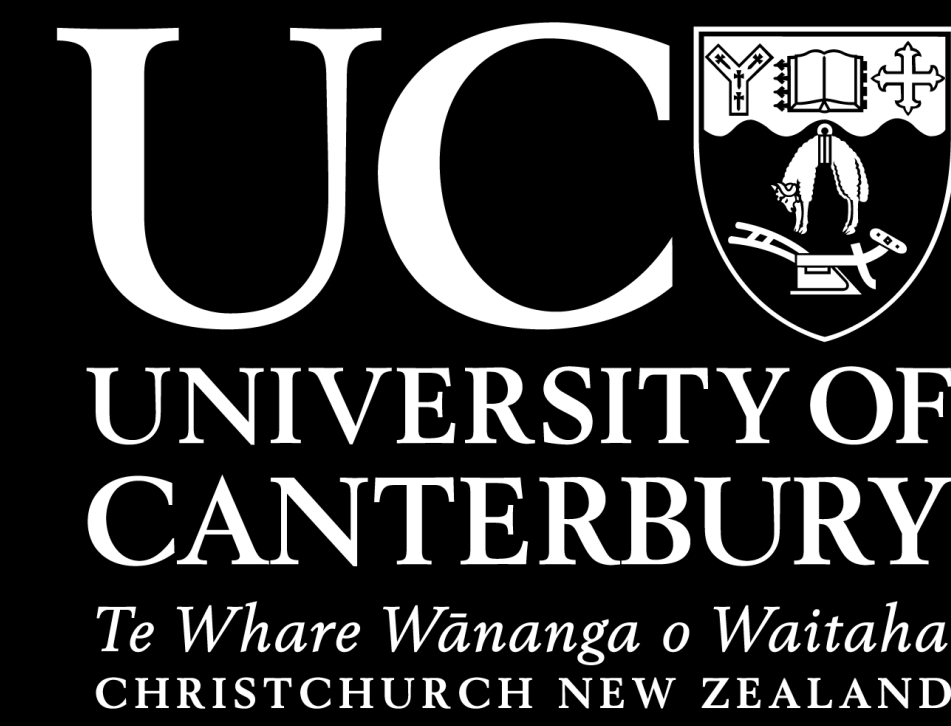


Scan to
download poster + references



Toward a Unified Database of Hybrid Broadband Ground-Motion Simulations for Historical Mw 3.5-7.8 Crustal and Subduction Earthquakes in New Zealand

Felipe Kuncar, Brendon Bradley, Robin Lee, Cesar Pajaro, Michael Dupuis, Jake Faulker, Sung Bae
University of Canterbury, Department of Civil and Environmental Engineering | felipe.kuncar@canterbury.ac.nz



References

New Zealand studies

Lee RL, Bradley BA, Stafford PJ, Graves RW, Rodriguez-Marek A (2022). Hybrid broadband ground motion simulation validation of small magnitude active shallow crustal earthquakes in New Zealand. *Earthquake Spectra*, 38(4): 2548–2579.

Dupuis MR, Lee RL, Bradley BA (2025a). Hybrid broadband ground-motion simulation validation of small-magnitude subduction earthquakes in New Zealand. *Earthquake Spectra*.

Kuncar F, Bradley BA, Lee RL, et al. (in progress). Validating hybrid broadband ground-motion simulation of moderate-magnitude active-shallow-crustal earthquakes in New Zealand.

Dupuis MR, Lee RL, Bradley BA (2025b). Hybrid broadband ground-motion simulation validation of moderate-magnitude subduction earthquakes in New Zealand. To be submitted.

Pajaro C, Bradley BA, Lee RL (2025). Validation of hybrid broadband ground-motion simulation for large magnitude earthquakes in New Zealand. New Zealand Society of Earthquake Engineering Annual Conference. 9 pp.

Global studies

Goulet CA, Abrahamson NA, Somerville PG, Wooddell KE (2015). The SCEC Broadband Platform validation exercise: methodology for code validation in the context of seismic hazard analyses. *Seismological Research Letters*, 86(1): 17–26.

Dreger DS, Beroza GC, Day SM, Goulet CA, Jordan TH, Spudich PA, Stewart JP (2015). Validation of the SCEC Broadband Platform V14.3 simulation methods using pseudospectral acceleration data. *Seismological Research Letters*, 86(1): 39–47.

Taborda R, Azizzadeh-Roodpish S, Khoshnevis N, Cheng K. (2016) Evaluation of the southern California seismic velocity models through simulation of recorded events. *Geophysical Journal International*, 205, 1342-1364.

Paolucci R, Smerzini C, Vanini M (2021). BB-SPEEDset:: a validated dataset of broadband near-source earthquake ground motions from 3D physics-based numerical simulations. *Bulletin of the Seismological Society of America*, 111, 2527-2545.

Graves R (2022) Using a grid-search approach to validate the Graves–Pitarka broadband simulation method. *Earth, Planets and Space*, 74(1): 186.

Sajan KC., Chukwuebuka N., Stewart JP, Graves RW (2024). Reconciling bias in moderate magnitude earthquake ground motions predicted by numerical simulations. Geo-Congress. 10 pp.

Pinilla-Ramos C, Pitarka A, McCallen M, Nakata R (2025). Performance evaluation of the USGS velocity model for the San Francisco Bay Area. *Earthquake Spectra*, 41(1): 457-494.

Sajan KC, Chukwuebuka N., Graves RW (2025). Frequency based validation of site response of physics-based simulation in Northern California. SCEC Report Project #21143. 17 pp.